

Checkpoint F3: Line integrals

Consider the vector field $\vec{F}(x,y) = \langle y^2, -xy \rangle$.

1. Use CalcPlot3D to draw a picture of this vector field. Include a screenshot.
2. Say an ant is walking along the quarter-circle of radius 2 **from** the point $(2,0)$ **to** the point $(0,2)$. Overall, is this vector field helpful or annoying to the ant?
3. Compute the line integral $\int_C \vec{F} \cdot d\vec{r}$, where C is the quarter-circle of radius 2 from the point $(0,2)$ to the point $(2,0)$.
(Careful: how is that different than the previous question?)